

Facial Video-Based Photoplethysmography to Detect HRV at Rest

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Key words

- photoplethysmography
- heart rate variability
- facial video signal
- cardio-respiratory fitness
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- computer vision

Abstract



Our aim is to demonstrate the usefulness of photoplethysmography (PPG) for analyzing heart rate variability (HRV) using a standard 5-min test at rest with paced breathing, comparing the results with real RR intervals and testing supine and sitting positions. Simultaneous recordings of R-R intervals were conducted with a Polar system and a non-contact PPG, based on facial video recording on 20 individuals. Data analysis and editing were performed with individually designated software for each instrument. Agreement on HRV parameters was assessed with concordance correlations, effect size from ANOVA and

Bland and Altman plots. For supine position, differences between video and Polar systems showed a small effect size in most HRV parameters. For sitting position, these differences showed a moderate effect size in most HRV parameters. A new procedure, based on the pixels that contained more heart beat information, is proposed for improving the signal-to-noise ratio in the PPG video signal. Results were acceptable in both positions but better in the supine position. Our approach could be relevant for applications that require monitoring of stress or cardio-respiratory health, such as effort/recuperation states in sports.

Introduction



Heart rate variability (HRV) analysis is a valuable, simple and non-invasive method for analyzing the continuous changes in the sympathetic-parasympathetic balance of the autonomic nervous system (ANS). Fluctuations in the interval between consecutive heartbeats (the RR interval) can be interpreted as a sign of healthy cardiac function [8]. HRV has proven useful in clinical practice (for example, regarding cardiovascular disease, hypertension, diabetic autonomic dysfunction and psychological disorders) [1], and in sports science for assessing current physical state [7, 18, 19], as well as for identifying the anaerobic threshold in athletes [3, 11]. In recent years, there has been significant progress in the development of inexpensive technologies for measuring RR data in daily life conditions. In this context, several studies have proven the validity of different heart rate devices, which are chest strap-based such as Polar heart rate monitors, for detecting RR intervals in various conditions [22–16]. Some studies report an equivalent performance when compared to conventional ECGs [14, 21, 34], and

research groups often use the Polar system as a source of RR intervals, which are subsequently analyzed using separate software [26]. Other studies provide data suggesting that beat-to-beat (BB) intervals obtained using the photoplethysmographic (PPG) technique could replace the RR intervals obtained from an electrocardiogram (ECG) and be used for HRV analyses [9], although the PPG technique seems to be subject to vascular variability. According to Poh et al. (2010), the ability to monitor a patient's physiological signals by remote, non-contact methods is a tantalizing prospect that would enhance the delivery of primary healthcare. Among such non-contact methods, PPG is a low-cost, non-invasive optical technique which measures arterial pulsations in the microvascular tissue bed through variations in transmitted or reflected light [2]. When surrogates of RR intervals are calculated from pulse wave signals (consecutive pulse-to-pulse intervals, PP), HRV is called pulse rate variability (PRV) by some authors [29, 35]. PPG is the most commonly used technique for PRV studies. For example, some studies have analyzed PRV extracted from the PPG signal through a fingertip attach-

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ment [25,28]. However, other studies recording PP intervals from PPG methods continue to use HRV when referring to both the RR and PP interval recording procedures [24]. We consider that the alternative use of HRV and PRV refers to the recording technique (RR or PP intervals, respectively), rather than to the subsequently performed calculations, which are always undertaken during the analysis of HRV parameters. Therefore, in our study, the whole procedure will be referred to as HRV, putting greater emphasis on the analytical system than on the recording of data. PPG using digital color video recordings of the human face have been shown to provide a valid and reliable method for assessing cardiac pulse and respiratory rates [23,32]. Advances in this non-contact technology present a great challenge in various fields such as telemedicine, home-based care, and other settings, including several sports conditions during training and competition. For example, Poh et al. [24] developed a method for automated computation of multiple physiological parameters (such as HR, RR and HRV), which represents one of the first demonstrations of a simple, low-cost method for non-contact HRV measurements. However, the video recording length was 1 min and therefore insufficient for drawing accurate conclusions about the HRV spectral parameters. It should be noted that, to extract comparable results, most research studies use recording periods of 5 min and force the breathing to 12 breaths per minute [22]. To the best of our knowledge, no studies have demonstrated the usefulness of this method for analyzing HRV using a standard 5-min test at rest with paced breathing, comparing the results with real RR intervals (not with the PP interval) and testing body positions other than sitting. Hence, the aim of the present study was to assess the practicability and the feasibility of a non-contact PPG method based on video recording of the human face for analyzing HRV, comparing the results for RR data and HRV parameters with those obtained simultaneously using a validated standard heart rate device. Data were obtained from a 5-min test at rest, pacing the breathing at 12 breaths/min in 2 conditions, the supine and sitting positions.

Method



Sample

Initially, the participants in the study were 23 Caucasian university students from 2 independent samples, each with different recording conditions, the supine position ($n=12$) and the sitting position ($n=11$). 3 participants, all from the sitting position group, were eliminated from the study due to artifacts in video recording (they moved too much during the video recording). Therefore, the final sample was composed of 20 participants: a) the supine position ($n=12$), with 2 women and 10 men (age: 25.75 ± 5.69 years); and b) the sitting position ($n=8$), with 3 women and 5 men (age: 25.62 ± 4.56 years). Participants did not take any kind of medication and did not show abnormal blood pressure or electrocardiographic patterns. All participants were volunteers and provided informed consent. The protocol was reviewed and approved by the local ethics committee and meets the Ethical Standards in Sport and Exercise Science Research [17].

Materials and procedure

The subjects' faces were recorded for 5 min using a compact camera (Canon Ixus 80is) with a resolution of 640×480 in AVI format. The camera was located at a distance of 80 cm from the face of the participant. It was attached to a special bar with an

adaptor that provides good stability, preventing movements that may affect recording. To control the effect of light on the video image, specific light conditions were established. The participant's face was only illuminated by a Sylvania Satin H7/H 230V 40W lamp located at the same distance as the camera. This lamp was chosen for its low amplitude fluctuations in illumination, which can create low-frequency components in the video signal. Simultaneously to the video recording, an inductive band (SleepSense, S.L.P. Ltd.) and a Polar chest strap (T61, Polar Electro Oy) were strapped around the chest of the subject to record the respiratory rate and the RR intervals. Special care was taken to guarantee a good electrical contact between the Polar chest strap electrodes and the skin chest of the participants to minimize noise and artifacts. The respiratory signal from the inductive band was amplified with a custom-made amplifier based on an INA116 (Analog Devices Inc.) and acquired with an A/D USB card (National Instruments, NI USB-6008). The RR signal from the Polar strap was acquired with an OEM receiver from the Polar chip (RE07S_ILNI_C), while the RR interval pulses were measured using a 16-bit timer from an 8-bit microcontroller (PIC18F1333, Arizona Microchip), with a clock frequency of 1 kHz (temporal resolution of 1 ms), and transferred via USB to a PC. Custom-made software developed in LabView was used to acquire and synchronize both signals (RR from Polar and the respiratory signal), as well as to record to a text file. The video files, RR intervals and respiratory rate signals were analyzed with MATLAB.

Data acquisition

Each participant wore the 2 elastic chest straps placed below the chest muscles for the recording of RR and breath data. Subsequently, participants had to undertake a 5 min HRV test at rest in an isolated and quiet room with controlled illumination. Participants lied on a stretcher (the supine group) or sat on a comfortable armchair (the sitting group) with eyes closed. They were asked to avoid making head or body movements and to synchronize their breathing to an electronic metronome (audio), operating at 12 cycles/min. A video of the full face was recorded during the 5-min HRV test and was saved in a non-compressed Red-Green-Blue (RGB) format. To ensure temporal synchronization, the RR intervals, respiratory rate and video recordings were started synchronously. Lighting conditions were kept constant throughout the recordings. The respiratory rate and RR intervals from the Polar strap were recorded using the Labview software.

HRV analysis

Prior to the HRV analysis, artifacts were identified and corrected with the Matlab software for the 2 RR signals (video and Polar). Hence, if the change from one RR interval to the next exceeded the interquartile range of the differentiated RR time series 10-fold, it was then considered that an artifact was present. The artifact was then classified as a missing beat (FN), extra beat or ectopic-like beat (FP). For a missing beat, the corrected RR interval was obtained by adding as many consecutive RR intervals as necessary to obtain a value close to the mean of the previous 10 RR intervals. An extra beat was split into as many RR intervals as necessary with an equal value, so as to be as close as possible to the mean of the previous ten RR intervals. An ectopic-like beat was substituted by 2 equal RR intervals, corresponding to the mean of the 2 RR intervals involved in the ectopic like beat. HRV analysis was performed using the Matlab software for the 2 RR signals, following the recommendations of the Task Force of the

European Society of Cardiology and the North American Society of Pacing and Electrophysiology [31]. For time domain analysis, the mean of RR intervals, the standard deviation of normal RR intervals (SDNN), the percentage of successive RR differences greater than 50 ms (pNN50), and the root-mean-square difference of successive normal RR intervals (RMSSD) were calculated for the 5 min period. Power frequency analysis of the 5 min recordings was performed sequentially with a fast Fourier transform based on a non-parametric algorithm with a Welch window after the ectopic-free data is detrended and resampled at 4 Hz with a cubic spline interpolation [22]. The power densities in the VLF band (0.00–0.04 Hz), the LF band (0.04–0.15 Hz) and the HF band (0.15–0.40 Hz) were calculated from each 5 min spectrum by integrating the spectral power density in the respective frequency bands. The maximum of the RR spectrum around the respiratory frequency (0.2 Hz) was located to identify the Respiratory Sinus Arrhythmia (RSA) video and Polar recordings. These peak frequencies were compared with the peak frequency of the respiratory signal spectrum recorded with the inductive band.

Computer vision analysis

1. Red-Green-Blue (RGB) decomposition from an AVI file. The analysis of the video signal was undertaken in 2 steps:

(1) The face of the subject was segmented manually by adjusting a rectangle to cover the maximum area of the face. While this part of the process was carried out manually, it could be easily automated using current computer vision-based facial feature detection algorithms [13]. The color video was stored with RGB color information for each pixel. 3 temporal signals with a sampling rate equal to the frame rate (30 Hz) were obtained. Several studies are recording between 15 and 30 frames per second to obtain RR intervals at rest from G video channel [23, 32]. The temporal resolution when detecting the pulse (heart beat) in our signal was 33.3 ms. This resolution ensured that all the heart beats are recorded during the 5 min rest. The cardiac signal was present in the 3 signals, with the G component having the highest amplitude. The G component was therefore used to obtain the cardiac information. The G channel signal was band-pass filtered between 0.4 Hz to 10 Hz with a zero-phase forward and reverse digital Butterworth second order filter to remove both low-frequency (respiration and movements) and high-frequency components (ambient light flickering, electronic noise in the optical sensor) (• Fig. 1).

(2) To improve the signal-to-noise ratio (SNR), only those pixels in the original video that had a high correlation coefficient with the previous G filtered signal were averaged. To accomplish this, each frame was divided into blocks of 20×20 pixels, and the G component was averaged for each block. The 32×24 signals obtained from each frame were cross-correlated with the previous G-filtered signal. Only the signal blocks that had a correlation coefficient higher than the specific value were averaged (for example, 0.7). • Fig. 2 shows an image of the correlation coefficient for each block for a single case, and it can be seen that some parts of the face (forehead, cheeks and mouth) had more heartbeat signals than other parts (nose, eyes).

2. Filtering of artifacts and RR calculation. HRV parameters (for example, LF and HF spectral components) are not calculated from the raw data of the G channel. These parameters are calculated from the RR series, which are calculated from the G

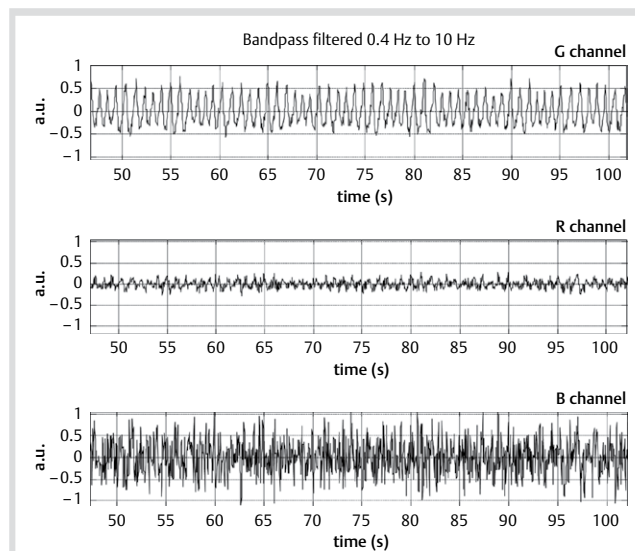


Fig. 1 G, R and B signal components of facial video (50 s), with mean removed and band-pass filtered (0.4 Hz to 10 Hz). The RR peaks in the green channel can be clearly seen.

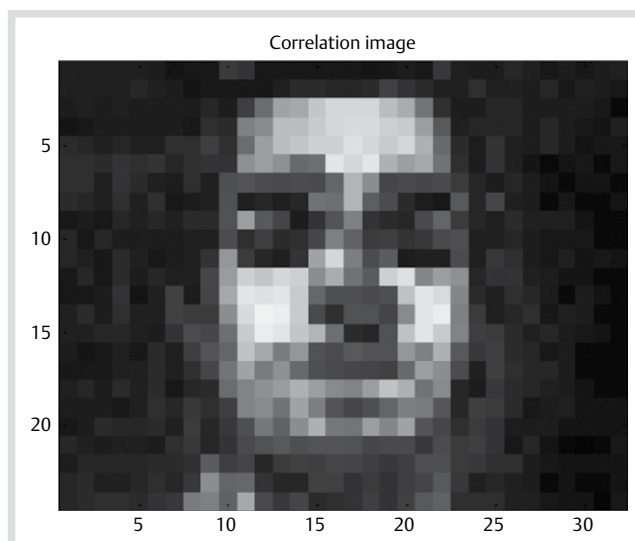


Fig. 2 Correlation coefficient for each image block with the reference 200 signal for a 5-min facial video. The white parts (forehead, cheeks and chin) have a greater cardiac component than other parts (nose, eyes).

channel. Thus, the G component obtained from the blocks with higher correlation was used to obtain the RR signal. In order to improve the temporal resolution on the detection of the beat position and have the same resolution as the Polar strap, the signal was resampled from 30 Hz to 1 kHz using a cubic spline, according to the method proposed by Daskalov and Chirstov [12]. Subsequently, the signal was band-pass filtered between 1 Hz to 3 Hz with a second-order Butterworth filter and compared with a threshold equal to 0.8 times its standard deviation. The local maxima in the resulting signal were used as estimates of the beat position. The RR series was obtained by differentiation of the beat positions.

3. Synchronization of the 2 RR sources. The RR signals from the video signal and the Polar strap were aligned automatically using the maximum of the cross-correlation between the 2 signals (• Fig. 3).

Statistical analysis

All statistical analyses were performed separately for the supine and sitting positions. Calculation of the HRV parameters and of the Bland-Altman plots was carried out with MATLAB. SPSS for Windows (v20, IBM) was used to perform standard statistical methods for comparing HRV indices. The concordance correlation coefficient was calculated using the web-based support of the National Institute of Water and Atmospheric Research (NIWA). A general linear model (GLM) was used to detect possible differences in HRV parameters between the Polar and video systems and to calculate the effect size (ES) accordingly. The difference was considered to be small when $ES \leq 0.2$, moderate when ES was around 0.5, and great when $ES > 0.8$ [10]. Means and standard deviations, bias and limits of agreement (LoA) for both systems were also calculated. Homogeneity of variance and

homoscedasticity were verified by Levene tests. The concordance correlation coefficient was used to evaluate reproducibility between the video and Polar systems [20]. The Bland-Altman plots of the corrected Polar and the video time series were obtained for the sitting and supine positions [4]. Finally, the concordance correlation coefficient was also used to assess the degree of relationship between the real breathing rate obtained with the inductive band, and the RSA obtained from the video and Polar systems.

Results

In total, 2 535 and 3 777 RR intervals were detected in the sitting and supine conditions, respectively. A total of 5 artifacts (2 FP and 3 FN) in the sitting condition and 42 artifacts (19 FP and 23 FN) in the supine condition were detected and corrected in the RR series from video recordings. No artifacts were present in the RR series from the Polar chest strap. **Table 1** shows there were no significant differences for the HRV parameters obtained from Polar and video systems in the supine position, except for SDNN, RMSSD, pNN50 and SD1. All parameters showed a small effect size (ES) except the 4 mentioned above, which only showed a moderate ES. However, for the sitting position, **Table 2** shows there were significant differences between the 2 systems regarding most HRV parameters obtained, except for LF, VLF, LF/HF, LFN and RSA. Most parameters showed a moderate ES, and only 2 parameters had a small ES (VLF and RSA). In addition to the data in the 2 tables, the real breathing rate obtained from the inductive chest strap was exactly 0.200 Hz ($SD = 0.0031$) for the supine position, and 0.201 Hz ($SD = 0.0013$) for the sitting position. These figures were exactly the same values obtained from the Polar system for both positions. There were also no significant differences for the video system: 0.200 Hz ($SD = 0.0033$; $ES = 0.017$) for the supine position and 0.201 Hz ($SD = 0.0016$; $ES = 0.125$) for the sitting position. Most concordance correlation coefficients for the comparison between the HRV parameters for

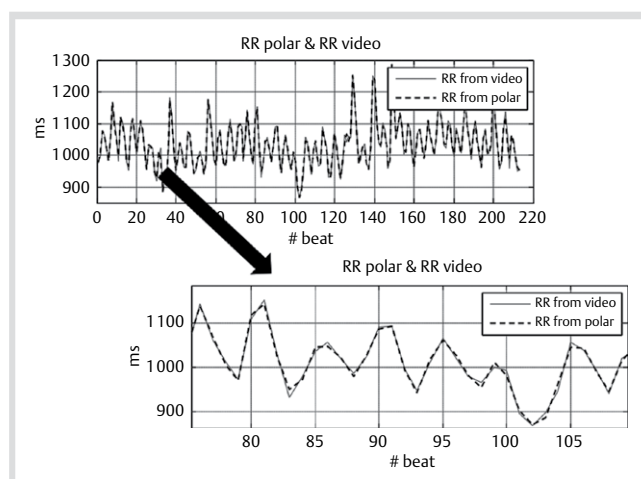


Fig. 3 Synchronization of the video and Polar signals for a single case (full recording and 35 s zoom). The figure shows the respiratory sinus arrhythmia (RSA) produced by the forced breathing at 0.2 Hz (12 peaks every 60 s).

Table 1 Heart rate variability parameters obtained from the Polar and video systems (means $\pm$ SD), correlation between Polar and video parameters, bias, magnitude of the bias, limit of agreement (LoA) and magnitude of the bias, for the supine position.

Parameter	Polar	Video	Correlation (pc)†	Bias	LoA		Magnitude of the bias	
					Inf.	Sup.	Effect size	Interpretation
RRmean (ms)	881.79 $\pm$ 199.48	881.47 $\pm$ 199.95	0.9999	-0.32	-1.77	1.14	0.021	small
SDNN (ms)	48.41 $\pm$ 19.14	52.51 $\pm$ 19.68 *	0.9544	4.12	1.51	6.72	0.524	moderate
RMSSD (ms)	41.43 $\pm$ 24.61	52.93 $\pm$ 22.88 * *	0.8398	11.5	6.59	16.41	0.707	moderate
pNN50	20.46 $\pm$ 22.03	29.13 $\pm$ 19.13 *	0.8635	8.66	4.38	12.95	0.643	moderate
VLF (m2)	596.57 $\pm$ 673.12	618.31 $\pm$ 706.97	0.9975	21.73	-5.27	48.73	0.222	small
LF (m2)	495.28 $\pm$ 615.11	502.77 $\pm$ 571.22	0.9849	7.49	-57.76	72.74	0.006	great
HF (m2)	1380.72 $\pm$ 1170.13	1453.98 $\pm$ 1108.28	0.9844	73.26	-44.86	191.38	0.145	small
LF/HF	0.45 $\pm$ 0.32	0.38 $\pm$ 0.28	0.8662	-0.07	-0.16	0.02	0.226	small
VLFn	28.09 $\pm$ 21.64	25.91 $\pm$ 17.83	0.9668	-2.18	-5.09	0.73	0.198	small
LFn	18.82 $\pm$ 11.36	18.29 $\pm$ 11.07	0.9752	-0.53	-2.08	1.01	0.050	small
HFn	53.09 $\pm$ 22.34	55.8 $\pm$ 18.12	0.9498	2.72	-1.06	6.49	0.185	small
SD1	29.3 $\pm$ 17.4	37.43 $\pm$ 16.18 * *	0.8398	8.13	4.66	11.6	0.707	moderate
SD2	61.34 $\pm$ 21.9	63.81 $\pm$ 23.22	0.9734	2.47	-0.42	5.35	0.244	small
RSA (breathing)	0.200 $\pm$ 0.003	0.200 $\pm$ 0.003	0.9115	0.0002	-0.0007	0.001	0.017	Small

Bias: Difference between Polar and video parameters; SDNN (ms), standard deviation of all RR intervals; RMSSD (ms), root mean square difference of successive normal RR intervals; pNN50, the percentage of successive RR differences greater than 50 ms; VLF (m2), very low frequency; LF (m2), low frequency; HF (m2), high frequency; LF/HF, LF/HF quotient; LFn, low frequency normalized; HFn, high frequency normalized; SD1 (ms) instantaneous beat-to-beat variability of the data; SD2 (ms) slow variability of heart rate; RSA, respiratory sinus arrhythmia

* $p < 0.05$; * * $p < 0.001$: levels of significance for comparison of the mean between the video and Polar systems

† Concordance correlation coefficient (pc). Strength-of-agreement: almost perfect $pc > 0.99$; substantial $pc > 0.95-0.99$; moderate $pc = 0.90-0.95$; poor $pc < 0.90$

the Polar and video systems were greater than 0.95 for the supine position. However, only 6 of the 14 coefficients were greater than 0.90 for the sitting position. The concordance correlation coefficient was 0.999 for the RR interval mean in both situations (► **Table 1,2**). The real breathing rate showed a maximum correlation with the Polar system for the supine position ($p=0.912$). ► **Fig. 4** shows the Bland-Altman plots for analyzing the relationship between the RR intervals recorded for the sitting and supine positions with the Polar and video systems. The bias in both cases was negligible (less than 0.45 ms). However, there was a small difference in the standard deviation of the differences between the sitting and supine positions (22.6 ms vs. 20.0 ms, respectively). The horizontal lines show the LoAs of the differences described in Bland and Altman (2007). A narrower LoA interval can be seen for the supine position than for the sitting position. In conclusion, results were acceptable in both positions but better in the supine position.

Discussion

Our study agrees with the importance of non-invasive measures of health and finds that modern computer vision techniques are a very promising tool towards this end. The results demonstrate

the feasibility of using facial video recordings at rest to obtain valid conclusions about HRV analysis, comparing the results to those obtained from a standard commercial device (Polar). In our study, better results were obtained in the supine position, for which the RR intervals and HRV parameters were more similar between the 2 systems than the data obtained for the sitting position. When the sitting position was compared with the supine position in greater detail regarding the data obtained, the supine position had higher concordance correlation coefficients for the comparison between the HRV parameters for the Polar and video systems (most above 0.9), as well as smaller differences between the means of those indices (► **Table 1,2**). The effect of RSA on heart rate was calculated from the spectral analysis of the RR intervals for the Polar and video records. Hence, the estimated breathing rate extracted from the 2 systems could be compared with the real breathing rate measured by the chest strap. Once again, for this parameter, a better concordance correlation between the video and Polar systems was also found for the supine position as compared to that for the sitting position. In all cases, the mean values were around 0.2, indicating that participants were correctly following the paced respiratory rate (12/min). According to Bland and Altman (1986), a high correlation does not automatically imply that there is good agreement between the 2 methods. Using the Bland-Altman representa-

Table 2 Heart rate variability parameters obtained from the Polar and video systems (means $\pm$ SD), correlation between Polar and video parameters, bias, magnitude of the bias, limit of agreement (LoA) and magnitude of the bias, for the sitting position.

Parameter	Polar	Video	Correlation (ρ) [†]	Bias	LoA		Magnitude of the bias	
					Inf.	Sup.	Effect size	Interpretation
RRmean (ms)	882.85 $\pm$ 52.35	883.23 $\pm$ 52.36 * *	0.9999	0.39	0.32	0.45	0.960	great
SDNN (ms)	40.90 $\pm$ 21.12	47.58 $\pm$ 18.71 *	0.9108	6.68	2.54	10.82	0.675	moderate
RMSSD (ms)	33.39 $\pm$ 22.77	52.69 $\pm$ 17.86 *	0.518	19.3	7.99	30.62	0.699	moderate
pNN50	14.1 $\pm$ 21.08	30.97 $\pm$ 16.13 *	0.5385	16.87	6.63	27.11	0.684	moderate
VLF (m2)	377.05 $\pm$ 256.44	382.01 $\pm$ 254.98	0.9979	4.96	-8	17.92	0.105	small
LF (m2)	210.82 $\pm$ 188.42	224.52 $\pm$ 176.31	0.9885	13.69	-5.91	33.3	0.280	moderate
HF (m2)	1215.21 $\pm$ 1992.77	1421.04 $\pm$ 1926.44 *	0.99	206.83	63.96	347.69	0.627	moderate
LF/HF	0.42 $\pm$ 0.43	0.23 $\pm$ 0.1	0.3186	-0.19	-0.48	0.1	0.257	moderate
VLFn	34.2 $\pm$ 22.18	27.55 $\pm$ 19.01 *	0.8928	-6.65	-12.34	-0.96	0.522	moderate
LFn	15.51 $\pm$ 9.48	12.94 $\pm$ 5.46	0.7934	-2.56	-6.18	1.06	0.285	moderate
HFn	50.3 $\pm$ 23.45	59.51 $\pm$ 17.68 *	0.7838	9.21	0.43	17.99	0.468	moderate
SD1	23.61 $\pm$ 16.1	37.26 $\pm$ 12.63 *	0.518	13.65	5.65	21.65	0.669	moderate
SD2	52.61 $\pm$ 25.62	55.73 $\pm$ 24.09 *	0.9858	3.12	0.99	5.25	0.631	moderate
RSA (breathing)	0.201 $\pm$ 0.001	0.201 $\pm$ 0.002	0.7094	0.0004	-0.0001	0.0013	0.125	small

Bias: Difference between Polar and video parameters; SDNN (ms), standard deviation of all RR intervals; RMSSD (ms), root mean square difference of successive normal RR intervals; pNN50, the percentage of successive RR differences greater than 50 ms; VLF (m2), very low frequency; LF (m2), low frequency; HF (m2), high frequency; LF/HF, LF/HF quotient; LFn, low frequency normalized; HFn, high frequency normalized; SD1 (ms) instantaneous beat-to-beat variability of the data; SD2 (ms) slow variability of heart rate; RSA, respiratory sinus arrhythmia. * $p < 0.05$; ** $p < 0.001$: levels of significance for comparison of the mean between the video and Polar systems

[†] Concordance correlation coefficient (ρ). Strength-of-agreement: almost perfect $\rho > 0.99$; substantial $\rho > 0.95-0.99$; moderate $\rho = 0.90-0.95$; poor $\rho < 0.90$

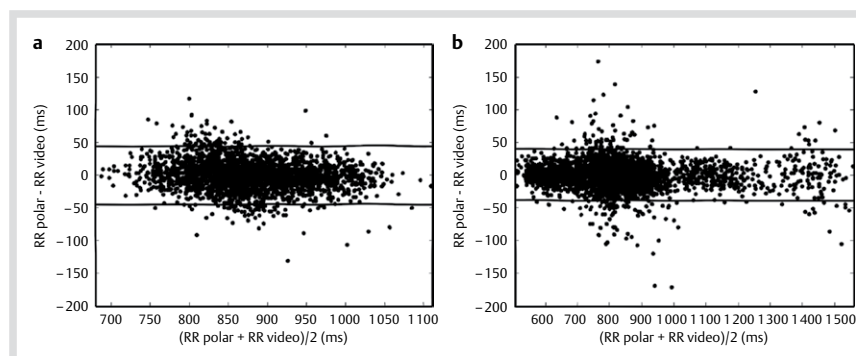


Fig. 4 Bland-Altman plot demonstrating the agreement between RR intervals from video and Polar systems (single subject). The lines represent the limits of agreement (LoA). X axis represents RR average for video and Polar systems. Y axis represents RR difference between Polar and video systems. **a** Results for the sitting position. **b** Results for the supine position.

tion, **Fig. 4** shows that there is good agreement between the video and Polar methods in the 2 different situations. However, the limits of agreement (LoA), determined by the space between the 2 horizontal lines in the figure, is narrower for the supine position, indicating a better agreement between the 2 recordings. No significant bias was introduced with the video method (less than 0.4 ms), as is also reflected in the RR means shown in **Table 1,2** (RRmean parameter). The values for the standard deviation of the differences may be smaller for the supine than for the sitting position (20.0 ms vs. 22.6 ms), which suggests that the results could be improved by reducing the RR outliers in video recordings, probably by having a better control of facial movements. One limitation of the sitting position is that head movements can increase the number of artifacts in the RGB signal, particularly in the G channel, which is the best channel for obtaining the RR signal. In addition, the movements produce changes in light brightness, which influence the filter of artifacts and calculation of the RR signal. In this context, it has been shown that the supine position is better for reducing these artifacts and for controlling the movements and changes of brightness. This is probably because, in this position, the head always rests against the stretcher surface. In contrast, in the sitting position, the participant has to maintain the effort required to keep the head upright for 5 min. 3 participants in the sitting position were excluded from the study because of the high density of artifacts in the RR intervals resulting from many head movements, which are easily identifiable when watching the video recordings. Therefore, the procedure should be improved by incorporating strategies to attenuate relatively small motion artifacts, especially in the sitting position [30]. In addition, improvements in the systems of facial tracking would also help to reduce the number of artifacts, thereby allowing data to be gathered in moving conditions (for example, stress tests). In this respect, it would be interesting to expand the research to other types of participants and situations, as PRV as an estimate of HRV has recently been shown to be sufficiently accurate only for healthy subjects at rest [27]. The practicability of the system requires further improvement. In particular, the recording and analysis of long periods during which all RR intervals can be studied would be an improvement over the current 5-min testing at rest.

A new procedure has been proposed for improving the signal-to-noise ratio in the PPG signal obtained from video recordings. Only those pixels that contained more heartbeat information (a higher correlation) were averaged. The results show that the forehead, cheeks and chin are the areas with more heartbeat signals. Another important aspect is that, in our study, an independent video camera was used with a sampling rate of 30 frames per second (FPS), which is required for capturing all heartbeats. The combination of these 2 factors allowed a rapid and objective analysis of the videos, when compared to other studies that have generally used either lower video sampling rates or more complex methods of video analysis [24,29]. Hence, our results demonstrate that the use of PRV derived from video PPG imaging can be a good indirect method for obtaining RR data and calculating the HRV parameters. Furthermore, other recent studies have successfully used PRV derived from digital PPG [25,29] and conclude that it could be used as a good surrogate for HRV [15]. The relevance of these studies is that they avoid the need for ECG to derive the RR intervals and respiration rate, allowing HRV parameters to be obtained just from the PPG signal. A study has suggested that the Polar system is not valid

for HRV analysis and that conventional ECGs should be used [33], but this was because the software included in the system used did not adequately correct the artifacts. Therefore, in accordance with other studies [26], we used a single software (based on MATLAB) that had been previously verified [16] for correcting the artifacts and for analyzing the HRV parameters of all of the recorded measurements of the 2 systems (video and Polar). In summary, the non-contact PPG method based on facial video recording at rest can be a good tool for indirectly detecting RR data for use in HRV. The results were better in the supine position, because the artifacts due to facial movement could be minimized. The advantage of such an approach is that current video cameras (such as webcams) are inexpensive, unobtrusive (do not require cables attached to the body) and available in a wide range of everyday devices, from laptops to smartphones. Better algorithms to compensate for facial movement and illumination changes would improve the results. Video-based HRV analysis could be relevant for applications that require ambulatory monitoring of cardio-respiratory health or stress induced by effort, especially in the supine position. For example, it could be used for the following: evaluating effort/recuperation states in sports science through a standard 5-min test at rest, such as for the prevention of overload/overtraining states, although it is advisable that the results are compared with those for other signs and symptoms of overreaching [6]; testing fitness and stress states in the general population at home; and remotely monitoring beat-to-beat recordings of patients in bed.

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